

Total No. of Questions : 8]

SEAT No. :

P-7626

[Total No. of Pages : 2

[6180]-146

T.E. (Information Technology)

DATA SCIENCE AND BIG DATA ANALYTICS

(2019 Pattern) (Semester - II) (314452)

Time : 2½ Hours]

[Max. Marks : 70

Instructions to the candidates:

- 1) All questions are compulsory.
- 2) Figures to the right indicate full marks.

Q1) a) Explain the process of reading and writing a file in HDFS with neat diagram. [8]

b) List and explain any four Hadoop shell commands with syntax. [4]

c) Differentiate between SQL and NoSQL databases with example. What is the need to develop big data applications using NoSQL databases? [6]

OR

Q2) a) What is the need of map reduce in Big Data? Explain the stages involved in map reduce task with a suitable example? [9]

b) Explain Hadoop ecosystem in detail. [9]

Q3) a) What is data wrangling? Why do you need it? Explain data wrangling methods? [9]

b) What is categorical variable? Why do you need categorical variable encoding? With an example, explain one-hot encoding. [9]

OR

Q4) a) Draw and explain Architecture of HIVE. [8]

b) How missing values and categorical variables are preprocesses before building model? Explain with suitable example. [4]

c) Explain z-score normalization. For the following dataset carry out z-score normalization (standardization), X = 23, 29, 52, 31, 45, 19, 18, 27. [6]

P.T.O.

- Q5)** a) What is Data Visualization? What are the major challenges in big data visualization and how to overcome these challenges? [6]
- b) Explain various techniques for visual data representation. [6]
- c) Explain the following data visualization techniques. [5]
- i) Candela
- ii) D3.js

OR

- Q6)** a) Explain data visualization with respect to 1-D, 2-D, 3-D data. [6]
- b) Explain various analytical techniques used in big data visualization. [6]
- c) Draw histogram with a suitable example and explain its usage. [5]
- Q7)** a) What is Porters valuation creation model? Explain porter's value chain analysis. [9]
- b) What is social media analytics? Explain the process of social media data analytics. [8]

OR

- Q8)** a) What is text mining? Draw and explain text mining architecture and its use. [8]
- b) Explain primary activities of Michal Porters value chain. [5]
- c) How mobile analytics is different from social media analytics? [4]

